Project - Low Level Design
On
Multimodal Manufacturing Creator

Course Name: Generative Artificial Intelligence

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

Sr no	Student Name	Enrolment Number
1.	SRASHTI SEN	EN22CS301983
2.	SOUMYA SHRIVASTAVA	EN22CS301978
3.	TANISHKA RAGHUWANSHI	EN22CS3011019
4.	TARUN PATIDAR	EN22CS3011034
5.	VIDHI PRAJAPATI	EN22CS3011076

Group Name: Group 12

Project Number: 12

Industry Mentor Name: Mr. Yash

University Mentor Name: Prof. Vineeta Rathore

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1. Introduction

The rapid advancement of **Generative Artificial Intelligence (GenAI)** has enabled systems that can generate multiple forms of content from a single input. In the manufacturing domain, visualization of product concepts plays a crucial role in design, prototyping, and presentation.

This project implements a **Multimodal GenAI application** that integrates **Large Language Models (LLMs)** with **Image Generation Models** through APIs to transform textual prompts into detailed textual explanations and visual representations of manufacturing concepts.

The system allows users to enter a prompt describing a manufacturing idea or product prototype. The application then generates:

- A **descriptive narrative** explaining the concept.
- A **high-quality image visualization** representing the product prototype.

This integration demonstrates the synergy between **text-to-text and text-to-image models**, enabling users to create **multi-dimensional content** suitable for educational demonstrations, research presentations, and product conceptualization.

1.1 Scope of the Document

This document provides the **Low Level Design (LLD)** of the Multimodal GenAI Application. It describes:

- Detailed system architecture
- Internal components and modules
- Data flow within the system
- Process workflow
- API integrations
- Design considerations

The document serves as a technical guide for developers, researchers, and system designers who need to understand the implementation and functionality of the system.

1.2 Intended Audience

This document is intended for the following stakeholders:

- **Software Developers** – For implementing and maintaining the application
- **System Architects** – For understanding the system structure and design decisions
- **Project Evaluators / Faculty Members** – For reviewing the project implementation
- **Researchers and Students** – For learning how multimodal AI systems are designed

1.3 System Overview

The **Multimodal GenAI Manufacturing Visualization System** is designed to convert **user-provided textual prompts into both descriptive text and product visualization images**.

The system integrates the following technologies:

- **Large Language Model (LLM)** for generating explanations and concept descriptions
- **Image Generation Model** for producing product prototype visuals
- **Backend Server** for handling API communication and logic
- **Frontend Interface** for user interaction

The workflow of the system includes:

1. User enters a manufacturing concept prompt.
2. The prompt is processed by the backend system.
3. The prompt is sent to an **LLM API** to generate descriptive content.
4. The same prompt is sent to an **Image Generation API** to generate a visual representation.
5. The system combines both outputs and displays them to the user.

This allows users to **visualize product concepts quickly without manual design tools.**

2. System Design

2.1 Application Design

The system follows a **three-tier architecture** consisting of:

1. Presentation Layer (Frontend)

Responsible for user interaction.

Functions:

- Accepts user prompts
- Displays generated text and images
- Provides a user-friendly interface

Technologies that can be used:

- Streamlit
- React
- HTML/CSS/JavaScript

2. Application Layer (Backend)

Handles system logic and communication with AI models.

Functions:

- Prompt processing
- API request handling
- Response formatting
- Error handling

Technologies:

- Python
- Flask / FastAPI

3. AI Model Layer

This layer integrates external AI services.

Includes:

- **Text Generation Model (LLM)**
- **Image Generation Model**

These models generate the outputs based on user prompts.

2.2 Process Flow

The operational workflow of the system is as follows:

1. User enters a manufacturing concept prompt.
2. The frontend sends the prompt to the backend server.
3. The backend processes the request.
4. The prompt is sent to the **LLM API** for narrative generation.
5. The prompt is also sent to the **Image Generation API**.
6. Both responses are received by the backend.
7. The backend combines and formats the outputs.
8. The results are displayed on the user interface.

2.3 Information Flow

The system follows a structured information flow:

User Prompt
↓
Frontend Interface
↓
Backend Processing Server
↓
LLM API (Text Generation)
↓
Image Generation API
↓
Backend Response Handling
↓
Frontend Display
↓
User Output

This ensures efficient communication between system modules.

2.4 Components Design

The system consists of the following components:

1. User Interface Module

Handles user input and displays generated results.

Features:

- Prompt input field
- Output display section
- Image visualization panel

2. Prompt Processing Module

Responsible for preparing prompts for AI models.

Functions:

- Prompt validation

- Input formatting
- Prompt optimization

3. Text Generation Module

Uses an **LLM API** to generate textual explanations of manufacturing concepts.

Output includes:

- Product description
- Functional explanation
- Use cases

4. Image Generation Module

Uses an **image generation API** to generate product concept images.

Output includes:

- Prototype visualization
- Concept illustrations

5. Response Integration Module

Combines text and image outputs before displaying them to the user.

Functions:

- Formatting output
- Error handling
- Response synchronization

2.5 Key Design Considerations

Several factors were considered while designing the system:

Scalability

The system should handle multiple user requests and support future integration of additional AI models.

Performance

Efficient API calls and response handling ensure faster output generation.

Usability

The interface should be simple and easy for users without technical knowledge.

Reliability

Proper error handling ensures the system functions even when APIs fail or network issues occur.

Security

API keys and credentials must be securely stored to prevent unauthorized access.

2.6 API Catalogue

The application integrates external APIs for AI processing.

1. Text Generation API (LLM)

Purpose:
Generate descriptive explanations for manufacturing concepts.

Input:
User prompt

Output:
Detailed textual narrative

Example Use Case:
Explaining the design and functionality of a smart manufacturing robot.

2. Image Generation API

Purpose:
Generate visual representations of product prototypes.

Input:
Text prompt describing the product concept

Output:
Generated image visualization

Example Use Case:
Generating a conceptual design of an automated assembly machine.

3. Backend Service API

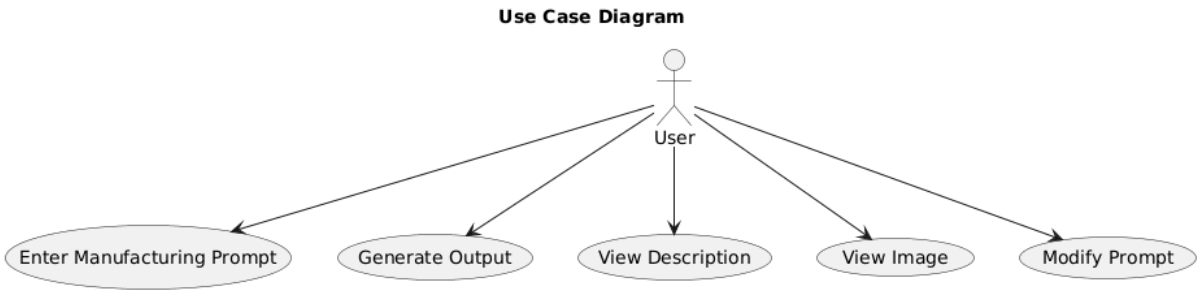
Purpose:
Manage communication between frontend and AI services.

Functions:

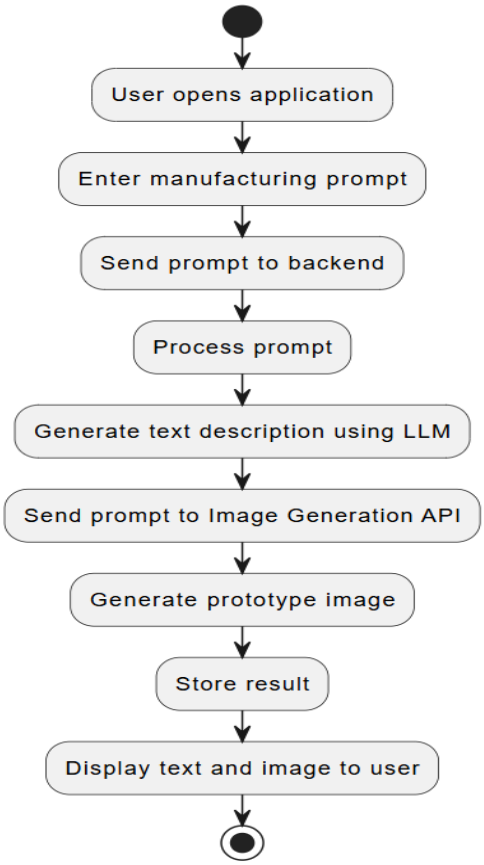
- Receive user input
- Send requests to AI APIs
- Return generated outputs

4.Architecture Diagram

Use Case Diagram



Activity Diagram



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